Meeting - Feb 19, 2026

February 17, 2026 · Thomas Haase

Summary

- Data: (User)–reply–>(User) network created from reply-trees below politicians tweets.
- Power-law structure indicates hierarchical network
- One giant component with 96% of all nodes
  - giant component is sparse and hub-driven (low clustering, mediocre reciprocity).
- More users send replies than receive them (outdegree > indegree), and receiving replies is far more concentrated

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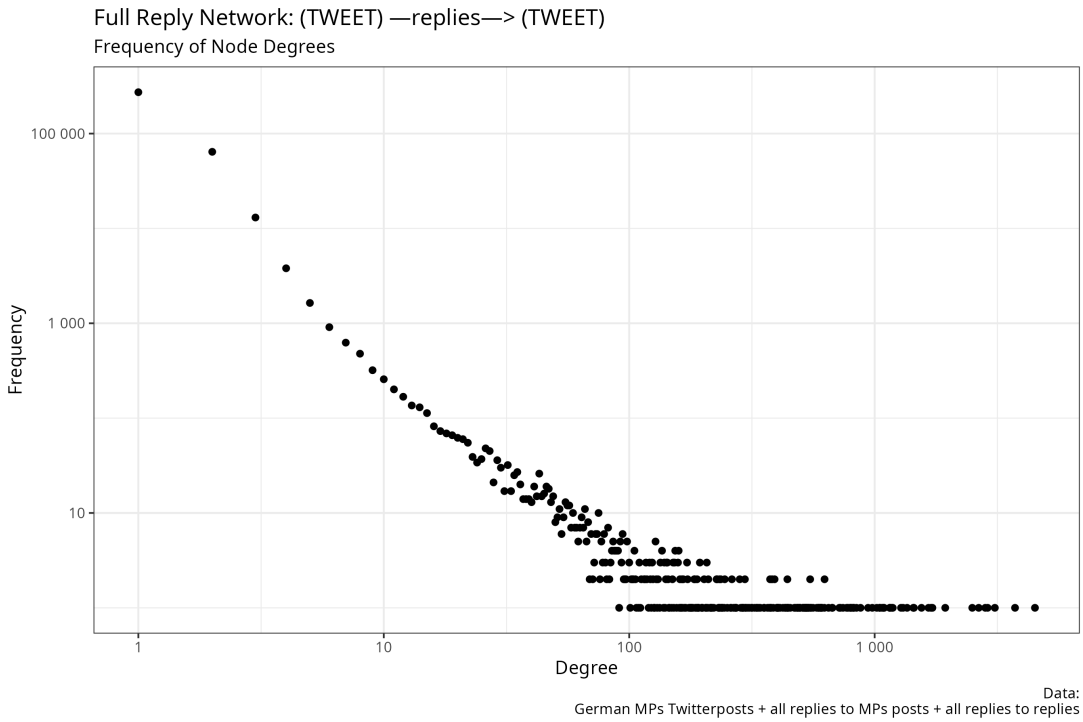
Data Collection Process

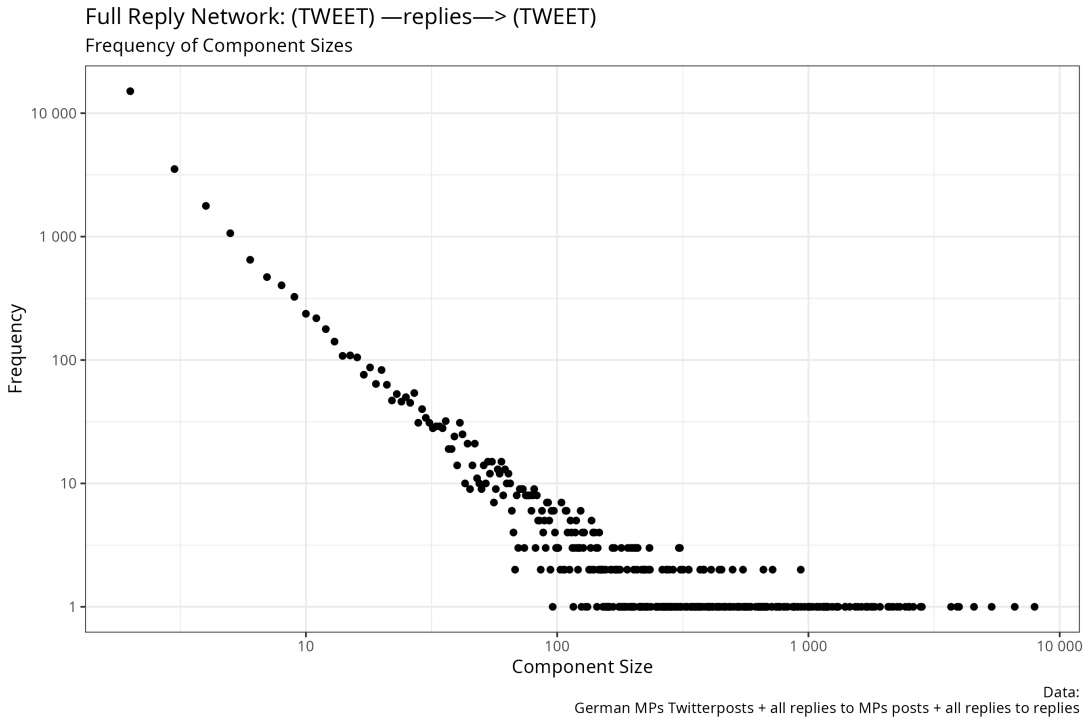
(Data Scraped by Armin Pournaki):

Data were collected via Twitter's streaming API using two parallel strategies to capture reply interactions involving German MPs. The follow stream tracked replies to MP-generated tweets. The track stream captured tweets mentioning MP handles (e.g., "@username"), including direct replies and nested replies to reconstruct complete conversation threads. Protected accounts were excluded. Both datasets were combined and deduplicated to create the final reply dataset.

Discussion Trees (Tweet —reply–> Tweet)

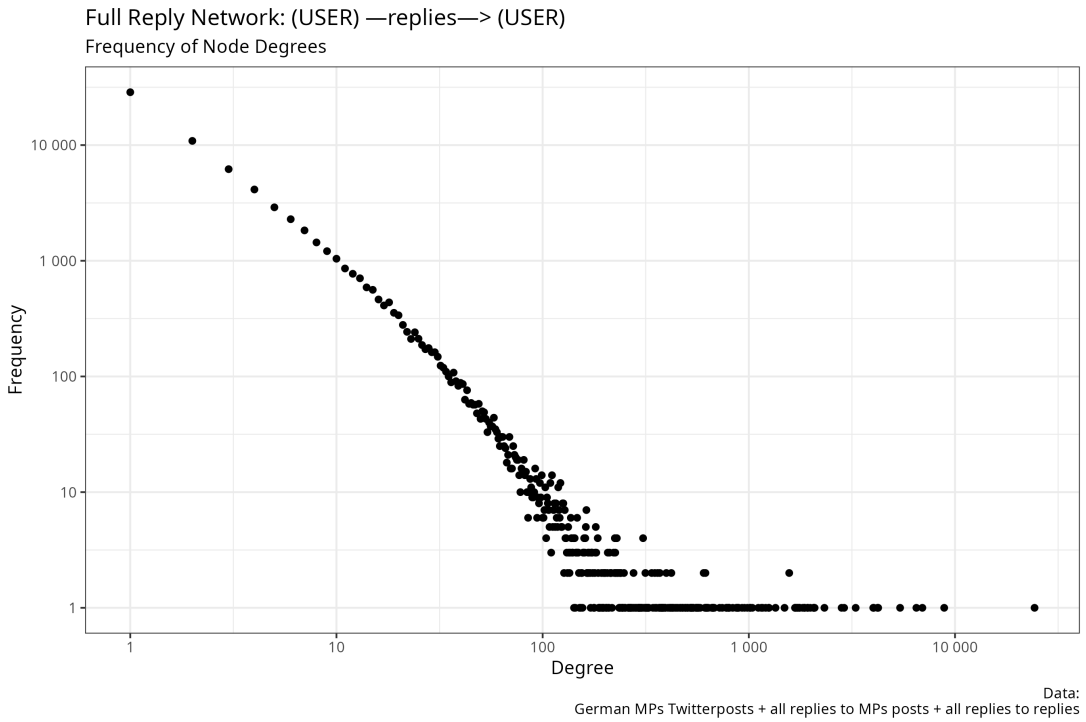


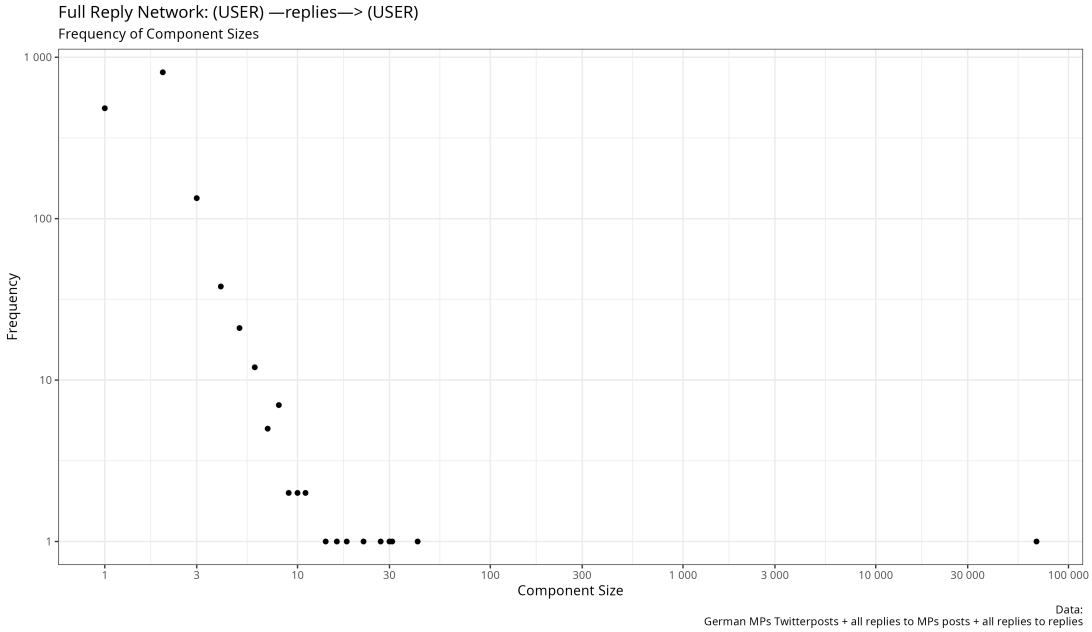
- Powerlaw
  - many tweets without/ with few replies
  - few tweets with a lot of replies



- Powerlaw
  - many discussionthreads with small size
  - few discussionthreads with a lot of replies

User-centered Network (User —reply–> User)





- There is one Giant Component

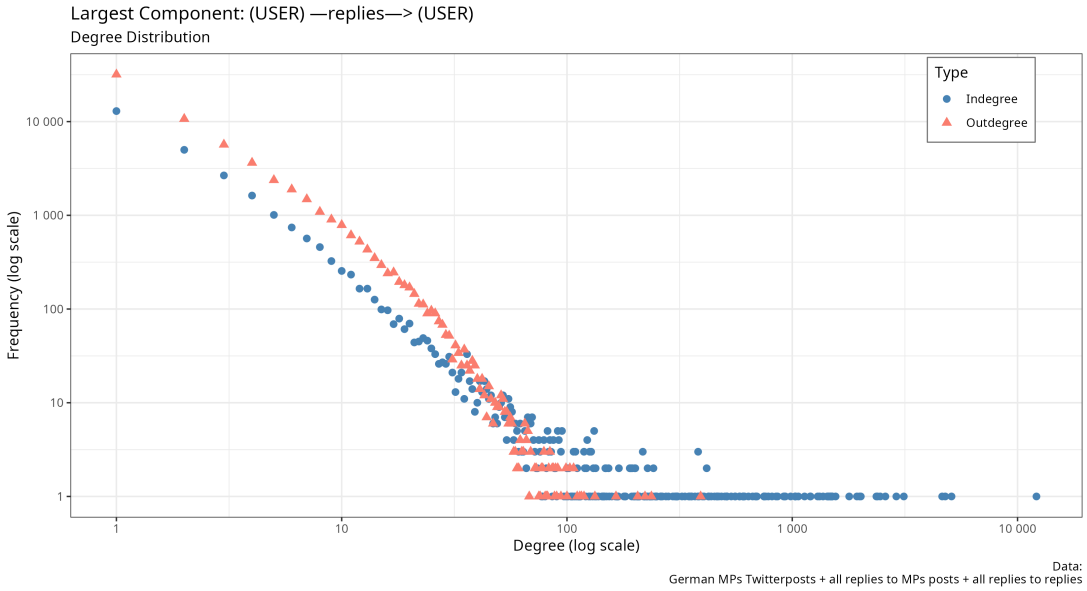
The giant component holds 95% of all nodes in the network

Component Size	Number of Components	Total Nodes	% of Network
68240	1	68240	95.55%
42	1	42	0.06%
31	1	31	0.04%
30	1	30	0.04%
27	1	27	0.04%
22	1	22	0.03%
18	1	18	0.03%
16	1	16	0.02%
14	1	14	0.02%
11	2	22	0.03%
10	2	20	0.03%
9	2	18	0.03%
8	7	56	0.08%
7	5	35	0.05%
6	12	72	0.10%
5	21	105	0.15%
4	38	152	0.21%
3	134	402	0.56%
2	807	1614	2.26%
1	483	483	0.68%

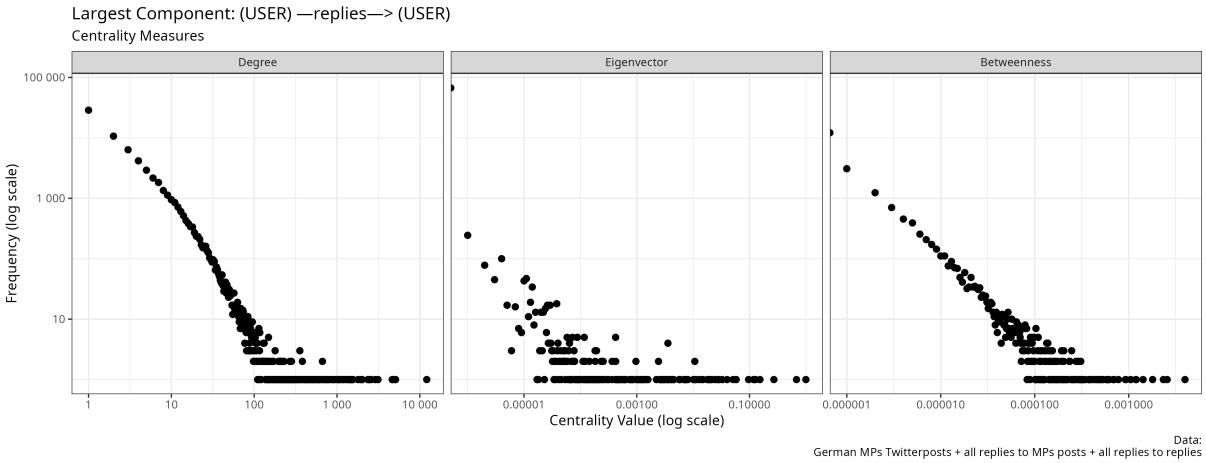
Largest Component

Metric	Value
Nodes	68240
Links	237264
Density	0.0001
Mean Degree	6.95
SD Degree	70.37
Reciprocity	0.13
Assortativity (Degree)	−0.10
Average Shortest Path (sampled)	4.35
Diameter (sampled)	47
Clustering Coefficient	0.1267
Max K-Core	24

- tiny density
  - sparse network
- large degree standard deviation
  - few super central nodes
- 13% of connections are mutual
  - few actual discussions, most interactions between “strangers”
  - People post to politicians root tweets, Politicians do not answer
- slightly disassortative
  - high-degree hubs connect to low-degree periphery nodes
- low short path rate (4.35)
  - lot of users probably connected through hubs
- little clustering (12%)
  - few smaller groups
- sampled k-core max 24
  - pretty well connected core



- Outdegree > Indegree
  - more users send a moderate number of replies than receive them
- Indegree has a longer tail than outdegree
  - some users receive replies from a lot of different people, but no single user is actively replying to thousands of distinct others. Receiving replies is more concentrated than sending them.



- Eigenvector
  - very compressed
    - most users are connected to low periphery users
  - few nodes with high centrality
    - suggesting a small, tight, influential core isolated from the rest